For instance, let us consider the matrix $A=\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$.

Suppose that we do the transformation $R_2 \to R_2 + \lambda R_3$ on A, where $\lambda \neq 0$ is a constant. Then, we get

$$A \xrightarrow{R_2 \to R_2 + \lambda R_3} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21}+\lambda a_{31} & a_{22}+\lambda a_{32} & a_{23}+\lambda a_{33} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}. \quad \ldots.(1)$$

The matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & \lambda \\ 0 & 0 & 1 \end{bmatrix}$ is an elementary matrix, since we have $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 \to R_2 + \lambda R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & \lambda \\ 0 & 0 & 1 \end{bmatrix}$.

Pre-multiplying A by $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & \lambda \\ 0 & 0 & 1 \end{bmatrix}$, we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & \lambda \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21}+\lambda a_{31} & a_{22}+\lambda a_{32} & a_{23}+\lambda a_{33} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}. \quad \ldots (2)$$

From (1) and (2), we get $A \xrightarrow{R_2 \to R_2 + \lambda R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & \lambda \\ 0 & 0 & 1 \end{bmatrix} A.$

So, the effect of applying the elementary transformation $R_2 \to R_2 + \lambda R_3$ on A is the same as that of pre-multiplying the matrix A with the elementary matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & \lambda \\ 0 & 0 & 1 \end{bmatrix}$.

Similarly, we can show that

(i) the effect of applying the elementary transformation $R_2 \leftrightarrow R_3$ on A is the same as that of pre-multiplying the matrix A with the elementary matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$.

(ii) the effect of applying the elementary transformation $R_2 \to \lambda R_2$ on A is the same as that of pre-multiplying the matrix A with the elementary matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

We state the following result without proof.

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